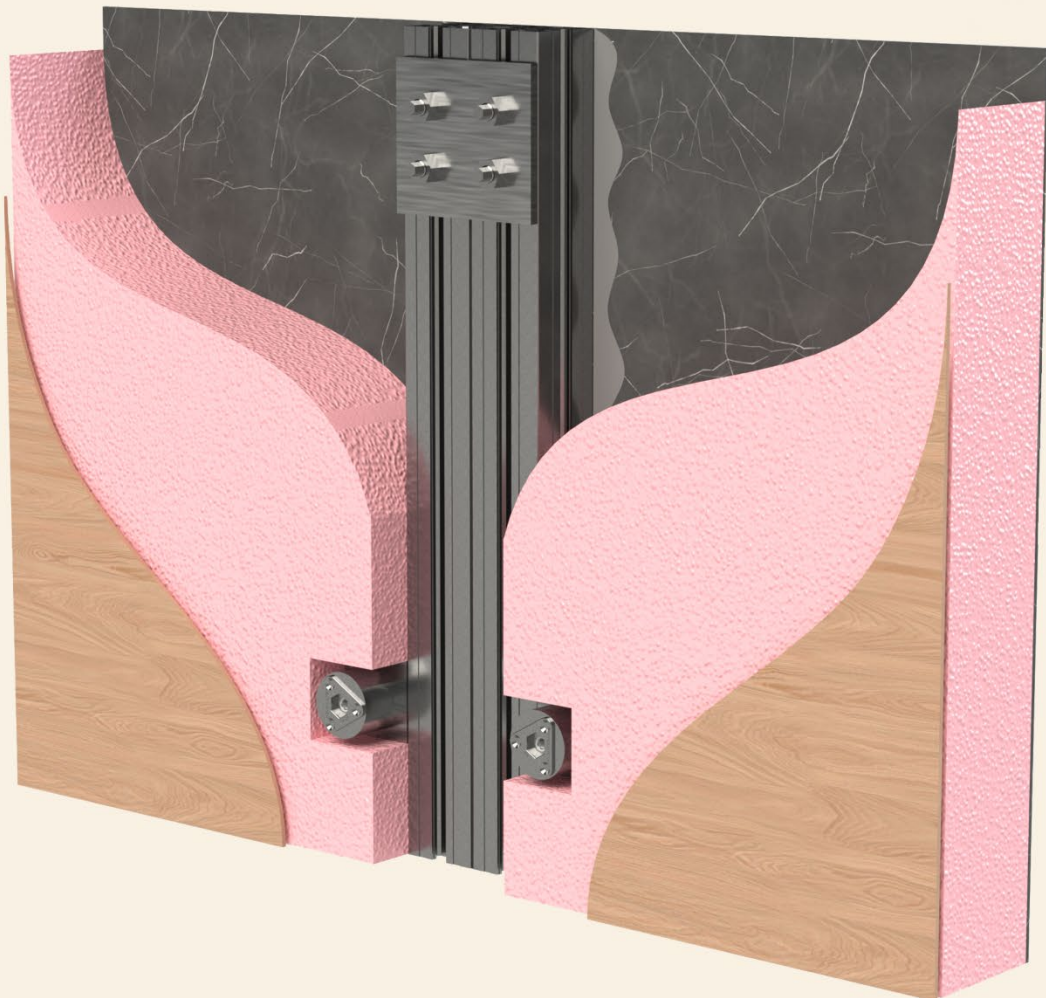




PROTEUS

System of Upgradable Architecture

Exergesis



Blake Reed 2025

Design and Evaluation of the Proteus Adaptive Wall System for Modular Housing in Australia

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Last, but not least, I am grateful to the **scholars and to the wider research fields** on which this project is based. This project is part of a bigger idea of redesigning our housing needs in a way that is versatile, sustainable, and towards the future.

Abstract

This exegesis is an exploration of the design research, development, and evaluation of the Proteus upgradable Architecture system, which is a modular aluminium wall framework targeting flexibility, repairability, and circular performance in Australian modular housing. The Proteus wall panel is situated within the topics of Design for Manufacture and Assembly (DfMA) and Design for Disassembly (DfD) in an attempt to address the historical insufficiencies of rigidity, and make use of the advantages of prefabricated housing via rapid panel interchangeability.

The research adopts a mixed-method approach involving literature synthesis, observations from Wild Modular construction sites, and a comprehensive series of iterative prototyping. The document discusses how the Proteus system can be developed using DfMA principles, modular detailing, and digital fabrication to make an adaptable, strong and weatherproof wall structure using standardised aluminium extrusions, service cavities, and compression seals.

This exegesis identifies that the Proteus system contributes to Australia's broader trend of circular and adaptive design practices in residential construction by designing to meet the intent of NCC 2022 Class 1 housing standards. It evaluates the potential and limitations of the design and suggests a model for integrating adaptable elements into modular systems. This exegesis concludes that design-led innovation can transform the housing industry into a sustainable model driven by materials intelligence, disassembly logic, and user focus.

Keywords: modular construction, adaptive architecture, DfMA, DfD, circular economy, aluminium design, sustainability

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Chapter 1: Introduction

1.1 Context and Background

Housing affordability is a critical problem currently faced by Australia, fostered by the growing price of materials, labour, land and the inherited culture of home size expansion . The detached home of the Australian average is now much too large for most households to be usefully filled. In other words, it imposes substantial financial hurdles for young couples and smaller families. By introducing the Proteus adaptive wall system, the situation can be overcome.

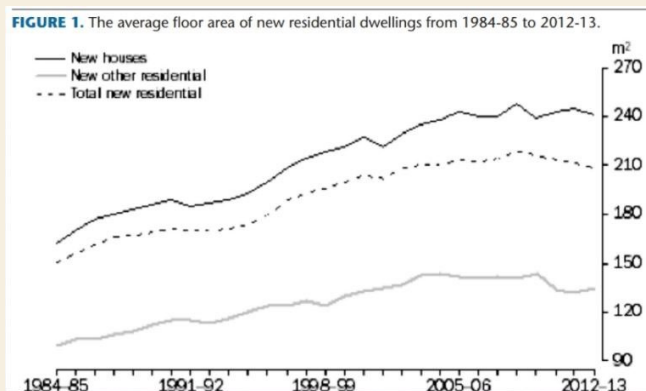


Figure 1.1: Graph showing increase in house floor size

The Proteus wall system overcomes this by modularity which is flexible and. This means that an occupant does not need to purchase a bigger and more fixed dwelling. Instead, they can invest in a modest module that offers the necessary living spaces and subsequently add bedrooms, closets, bathrooms, or offices as their family or demand increases . Furthermore the design makes use of Screwpier foundations, which replaces concrete, and can be removed. This is consistent with the ideas of sustainable consumption and efficient resource utilisation. As a result, Proteus' building components pair architectural scalability with ease of assembly/disassembly: screw-pier foundations and lightweight aluminium framing make fast assembly and disassembly without concrete or deep excavation possible, whilst allowing for full removability if necessary.



Figure 1.2: Illustration showing screw piers used as f a house foundation

1.2 Research Problem

In Australia, traditional modular housing is often limited in its adaptability by fixed component fasteners; nor do these homes allow for post-installation flexibility. Most systems are designed for assembly efficiency, not ongoing modification. Thus, when it does come to replace or upgrade, the process is expensive and wasteful (Crowther 2022; Rahman 2014). Therefore, there is a need for innovation in design that will permit disassembly, repair, and expandability while still complying with regulations, engineering capability, circular economy and being environmentally friendly.

The Proteus adaptive wall system was designed to tackle this problem. It makes use of an aluminium extrusion-framed modular wall which incorporates removable aluminium cladding panels, insulated service cavities and compression seals – being able to allow for mechanical attachment, weatherproofing plus interchangeable parts without undermining the strength of the structure itself.

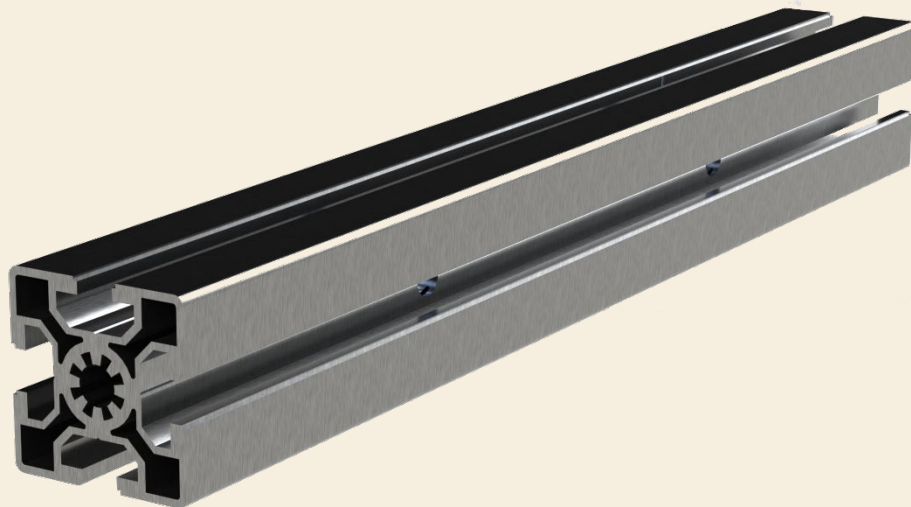


Figure 1.3: European Standard 50 series Aluminium T Slot Extrusion

1.3 Research Aim and Objectives

The main aim is to design, prototype, and evaluate an adaptive wall system that embodies the principles of Design for Manufacture and Assembly (DfMA) and Design for Disassembly (DfD) within the context of modular housing, to assist in the Australian housing crisis.

Specific objectives are to:

- Critically review existing literature on modular housing, DfMA, and DfD, with emphasis on Australian standards and construction practices.
 - Develop a conceptual and technical design for an adaptive wall system based on aluminium T-slot extrusions.
 - Prototype and test the system's assembly, accessibility, and upgradability
 - Evaluate its potential contribution to sustainable and circular housing models.
-

1.4 Significance of the Study

The research relates to the field of industrial design by providing a tangible demonstration of how product design thinking can address systemic inefficiencies in current established systems. By translating DfMA and DfD principles into architectural detail, the Proteus system bridges the gap between industrial design, construction innovation, and environmental sustainability (Kamali & Hewage 2016; Kibert 2016).

For Australia, where the Green Building Council of Australia (GBCA, 2024) and ABCB (2022) frameworks emphasise life-cycle efficiency and material reuse, this project presents a case study in integrating circular-economy thinking at the component scale. The work also aligns with UN Sustainable Development Goals 9, 11, 12, and 13, which call for resilient infrastructure, sustainable cities, responsible consumption, and climate action (UN 2015).

1.5 Significance Scope

The significance of the Proteus system lies in demonstrating that architectural flexibility and circular construction principles can co-exist with affordability and durability. While the current prototype targets low-rise residential applications, its engineering logic can scale to multi-residential or high-rise façade systems as it was designed to meet the intent of *AS/NZS 1170 Structural Design Actions* and *AS 4284 Testing of Building Façades* (ABCB 2022). These tests will be planned in the future of the project.

The project also contributes to emergency and temporary housing research: because the system relies on screw-pier foundations—large, galvanised steel screws installed several metres into the ground until stable foundation is reached—modules can be positioned quickly using hydraulic lifters off trucks or cranes, then removed or reconfigured without site damage. This makes the system suitable for disaster relief operations and some other remote situations where rapid deployment and de-mount ability are essential.

In summary Proteus provides a scalable, sustainable housing model which can adapt to the user's life cycle: starting off small, growing as necessary and staying portable throughout its entire lifetime.

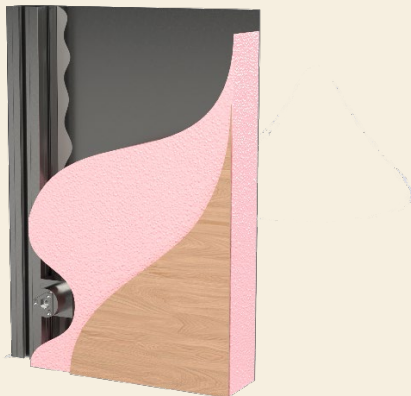


Figure 1.4: Conceptual overview of the Proteus adaptive wall framework



Figure 1.5: Example of modular housing installation observed at Wild Modular site

Chapter 2: Literature Review

2.1 Overview of Modular Construction and Modern Methods of Construction (MMC)

Research into modular construction consistently shows reductions in build time, waste, and embodied carbon compared with traditional construction (Ferdous et al 2019; Davis et al 2019). In Australia, off-site manufacture is also seen as a pathway to address housing shortages while maintaining quality control.

A key enabling technology for modular housing is the **screw-pier foundation system**, which replaces conventional concrete slabs with removable steel piers. These piers are screwed into the ground until they reach stable subbase, providing immediate load-bearing capacity without curing time or excavation. Once aligned, prefabricated modules are placed on top using hydraulic lifters or cranes, secured with mechanical brackets, and levelled before connection. Such systems are already used in emergency structures and relocatable housing, offering circularity benefits by allowing entire modules to be detached and reused elsewhere.

This technique directly informed the Proteus concept, which prioritises disassembly and minimal site impact as key sustainability outcomes.

2.2 Modern Methods of Construction and Prefabrication

Modern Methods of Construction (MMC) encompasses prefabricated, modular, and off-site fabrication strategies that shift labour from site to factory (with possibility of automation) to improve quality and reduce time, waste and cost (Arashpour et al 2023). According to BCA (2019), the integration of prefabricated prefinished volumetric construction (PPVC) within DfMA processes has been critical to Singapore's efficiency and safety gains—a precedent now informing Australian policy.

International studies confirm the environmental and productivity benefits of modularisation. For instance, Kamali and Hewage (2016) found that modular projects reduce embodied energy by up to 45 per cent compared with conventional construction, while Dong et al (2015) demonstrated a 25 per cent reduction in lifecycle energy use through factory-based assembly. However, Schweizer et al (2022) caution that results vary by supply-chain logistics, material selection, and end-of-life management.

In Australia, adoption remains slow due to fragmented supply chains, risk-averse procurement, and inconsistent regulation (Davis et al 2019; Manuel et al 2021). Nonetheless, Infrastructure Australia (2023) notes growing investment in volumetric and panelised systems for remote and social-housing projects, indicating a transition from pilot to mainstream deployment.

2.3 Design for Manufacture and Assembly (DfMA)

DfMA emphasises standardisation, tolerance control, and the simplification of component interfaces to enhance manufacturability (Gibb & Isack 2003; ABCB 2019). When applied to architecture, it promotes cross-disciplinary collaboration between designers, engineers, and manufacturers (Arashpour et al 2023). DfMA also aligns with lean-construction principles by eliminating non-value-adding tasks and reducing rework (Fard et al 2017; Duncheva & Hamzeh 2019).

The Building and Construction Authority (2019) highlights DfMA as a core mechanism for productivity improvement, encouraging modular kit-of-parts logic and early digital coordination.

For the Proteus wall, DfMA principles are evident in the consistent 50 × 50 mm aluminium T-slot module, pre-machined corner brackets, and bolt-on panel interfaces that reduce bespoke fabrication. As Lawson et al (2014) and Taylor (2010) argue, such modular standardisation enables scaling and localisation within varying climatic and regulatory environments.

2.4 Design for Disassembly (DfD) and Circular Construction

Aluminium has emerged as a preferred material for modular building systems due to its strength-to-weight ratio, corrosion resistance, and recyclability (Lawson et al 2014; Bai et al 2020). The Proteus system applies 50×50 mm European-standard T-slot aluminium extrusions, which enable structural connections using mechanical fasteners instead of welding—simplifying assembly and improving future disassembly.

Future iterations include custom aluminium extrusions for double-glazed window and door frames, maintaining continuity within the modular logic while enabling higher thermal and acoustic performance. These extrusions will integrate with the same T-slot system, ensuring compatibility between wall, door, and façade elements.

Various panel finishes, such as timber, gunmetal and full colours were analysed for texture, reflectivity, and sturdiness, indicating that aluminium cladding can provide aesthetic versatility.

2.5 Sustainability and Circular-Economy Frameworks

The built environment accounts for approximately 39 per cent of global CO₂ emissions (IEA 2019). Australia's commitment to net-zero by 2050 (DCCEEW 2023) places pressure on designers to integrate life cycle thinking into materials and processes.

(Kibert 2016) defines sustainable construction as balancing environmental, economic, and social outcomes across the building life cycle. (Kamali and Hewage 2016) and (Han et al 2020) highlight modular construction's potential to reduce embodied energy and improve resource efficiency through closed-loop systems.

Circular-economy principles articulated by (Ellen MacArthur Foundation 2019) and (UNEP 2020) advocate for product-service systems in which buildings act as “material banks.” In the Australian context, (GBCA 2024) encourages the specification of recyclable materials such as aluminium and the documentation of disassembly pathways to earn Green Star credits. The Proteus system aligns with these guidelines by enabling panel replacement and component recycling rather than demolition—a key contribution to SDG 12: Responsible Consumption and Production (UN 2015).

2.6 Material Innovation and Aluminium Applications

Material selection is central to modular system performance. Aluminium offers a favourable strength-to-weight ratio, corrosion resistance, and near-infinite recyclability (Australian Aluminium Council 2022). Although its production is energy-intensive, closed-loop recycling reduces embodied energy by over 90 per cent compared with

virgin smelting (Schweizer et al 2022). (Manalo et al 2016) and (Bai et al 2020) emphasise aluminium's durability in hybrid sandwich-panel construction, supporting longer service life and reduced maintenance.

In modular contexts, extruded-profile systems facilitate precision manufacturing and adaptable detailing (Ferdous et al 2021). (Lu et al 2020) and (Durmisevic 2019) further demonstrate that mechanical fasteners rather than adhesives enable reversible assembly, improving recyclability. The Proteus prototype design leverages these insights through a European Standard 50-series T-slot extrusion frame and custom stainless steel insertable bolts that can be replaced without end-feeding, promoting long-term reparability.



Figure 2.1: Custom Insertable Bolt Render



Figure 2.2: Custom CNC Machined Insertable Bolt

2.7 Policy and Regulatory Frameworks

The regulatory landscape greatly determines whether modular housing can succeed in Australia. The performance requirements for Class 1 buildings cover structural adequacy, fire safety, weatherproofing and energy utilisation. They are all laid out in detail throughout more than 500 pages of the National Construction Code 2022 (NCC

2022), which is presided over by the Australian Building Codes Board (ABCB 2022). What's more, the latest updates to Section J put fresh emphasis on very demanding standards for the thermal envelope and air tightness that are in line with those of Green Star Buildings v2 (GBCA 2024).

Complementary state initiatives such as the NSW Housing Strategy 2041 (NSW Government 2022) and Sustainability Victoria (2020) guidelines advocate for modular prefabrication to accelerate affordable housing while meeting environmental goals. Even with such frameworks in place, (Davis et al 2019) claim that policy fragmentation persists between planning certification manufacturing authorities. (Steinhardt and Manley 2016) writes, there are long-term good reasons for this division; yet others suggest that it should be overcome--for example (Van Gulijk, Tommerhope et al 2012) argue that harmonising building regulation with incentives for innovation is essential if MMC is to go mainstream as both a construction technique and legal category.

2.8 Critical Gaps in the Literature

While extensive research exists on modular efficiency and sustainability, several gaps remain pertinent to this study:

1. Limited empirical evaluation of component-level adaptability and long-term maintenance in Australian contexts (Manuel et al 2021; Crowther 2022).

2. Insufficient cross-disciplinary integration between industrial design and construction engineering—a space the Proteus project explicitly occupies.
3. Few detailed studies on aluminium-based modular systems for housing that combine DfMA and DfD principles.
4. A lack of comparative data on weatherproofing and service-access performance of demountable panels.

Addressing these gaps, the Proteus research aims to demonstrate how design-led experimentation can complement empirical building science to create more adaptable, circular, and regulation-compliant housing systems.

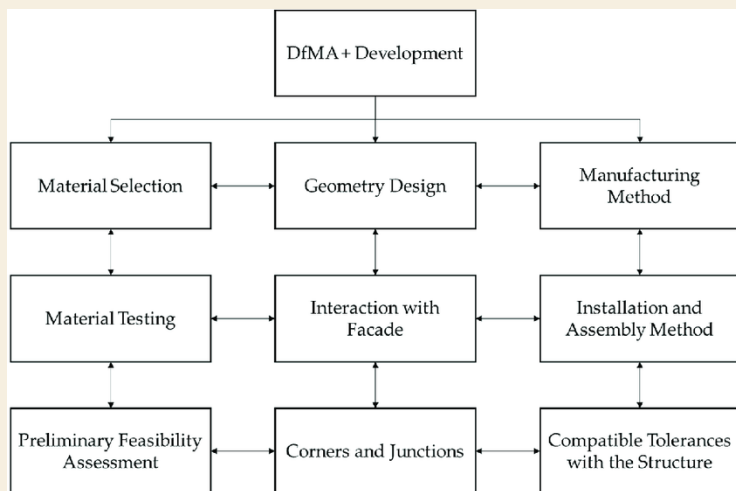


Figure 2.3: Framework of DfMA + DfD principles in modular construction

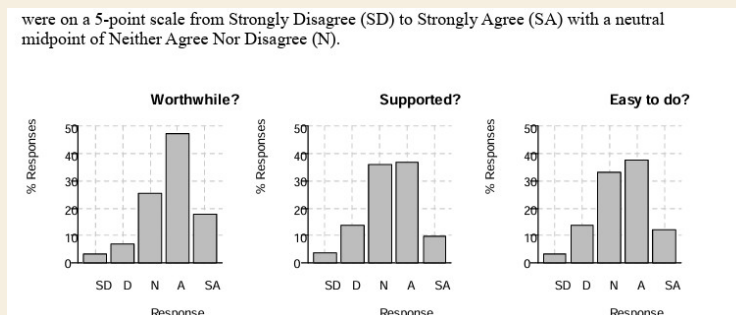


Figure 2.4: Construction workers perception of prefabrication in Australia (Blismas & Wakefield 2009, p. 23)

Chapter 3: Methodology

3.1 Research Design and Approach

This research adopts a *design-led mixed-methods* approach, combining qualitative analysis, case study observation, and iterative prototyping to evaluate the design, fabrication, and performance potential of the Proteus adaptive wall system. The methodology follows the “research through design” framework described by (Zimmerman, Forlizzi and Evenson 2010), in which designing and making become processes of inquiry that generate new knowledge.

According to (Godin and Beveridge 2021), design research involves “the exploration of form, function, and context through critical making,” enabling designs to embody hypotheses and reveal insights unattainable through theory alone. In this project, the wall system functions as both a product and a prototype—serving as an experimental apparatus for testing ideas about adaptability, modularity, and circularity.

The research therefore integrates:

1. *Theoretical exploration* — synthesis of MMC, DfMA, and DfD frameworks;
2. *Empirical observation* — documentation of modular housing installation practices through field visits with Wild Modular; Conducted over a 17 week internship (2 June – 9 October 2025) and
3. *In-Industry Practice* – Cad Modelling, Engineering Drawings, Research on currently existing solutions and communication with other field within the industry.
4. *Iterative prototyping* — creation, testing, and refinement of physical and digital models in response to design criteria.

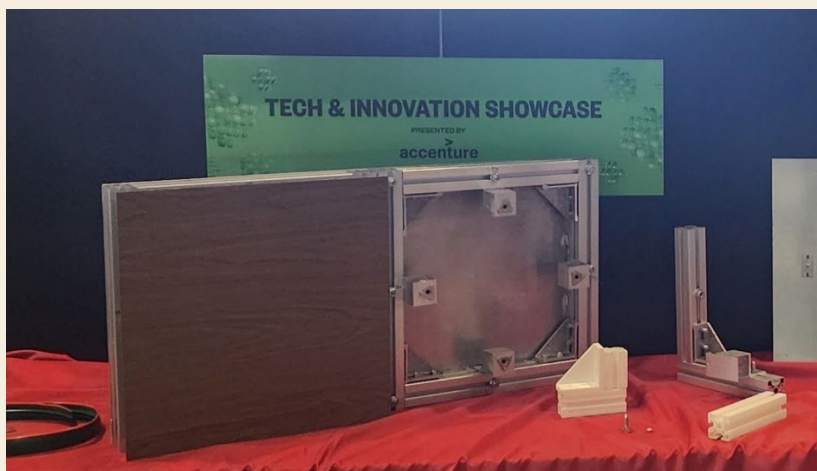


Figure 3.1: Proteus prototype, displayed at SXSW Sydney, Tech and innovation showcase 2025

3.2 Research Questions

1. How can Design for Manufacture and Assembly (DfMA) and Design for Disassembly (DfD) principles be applied to modular wall systems to improve adaptability and circularity in Australian housing?
2. What material and connection strategies enable weatherproof, demountable, and service-accessible wall construction without compromising structural performance or compliance?
3. How can industrial design contribute to bridging the divide between product-level detailing and architectural systems integration?

These questions align with the broader goals of Modern Methods of Construction and GBCA's (2024) sustainability framework, which both call for innovation at the component scale to advance systemic change.

3.3 Observations

An observation protocol represents site visits and office-based activities, ensuring systematic documentation of industrial processes and user needs. Notes and occasional photographs were logged daily. Insights gained were labelled under categories such. All processes noted in *Process Diary (2025)*.

Date / Period	Location	Activity	Data Collected
1 April – 17 June	University	Research, Initial sketches, Cad Prototyping and first scale prototypes	Research logs and digital prototypes
2 Jun – 9 Oct 2025	Wild Modular office, Pyrmont NSW	CAD, engineering drawings, and rendering	Digital models, time logs. Industry Experience
22 Jul – 5 Aug 2025	Home and University workshop	Testing of prototype framing, connection detailing	Assembly notes
6 Aug 2025	Site 1 – Wild Modular social housing	Observation of finished modular home, finishes, construction materials, etc	Field notes, annotated photos
6 Aug 2025	Site 2 – Wild Modular social housing	Observation of modular transportation, screw-pier installation, hydraulic lifters, alignment, installation, tolerance, and construction	Field photographs
6 Sep – 13 Oct 2025	Wild Modular design office & university/Home workshop	1:1 Prototype completion and presentation at SXSW Sydney 2025.	Photos, Verbal Communication, Physical testing.

3.4 Design Research Approach

The research employed a mixed-method design integrating CAD development, prototyping, and field observations through a professional internship with **Wild Modular** in Sydney. Over 352 hours, construction processes were documented across two modular-housing sites where prefabricated modules were transported on trailers, lifted by hydraulic lifters, and installed onto screw-pier foundations.

Direct observation and photographic evidence revealed that each pier penetrated several metres into the ground, substituting for concrete footings and allowing full module removal without demolition, whilst allowing for the possibility of easy expansion. These insights informed the development of Proteus connection details and lifting configurations.

Complementing field study, a **1:12 scale physical prototype** was fabricated using magnets to simulate fastener behaviour. This model incorporated interchangeable wall, door, and window panels, allowing iterative testing of layout, assembly sequence, and aesthetic finishes. The prototype validated the feasibility of modular expansion and formed the basis for later full-scale digital modelling and fabrication trials.



Figure 3.2: 1:12 House prototype, displayed at SXSW Sydney, Tech and innovation showcase 2025

3.5 Design Research Framework

The project follows a cyclical design research process comprising four stages, adapted from (Frayling 1993) and (Zimmerman et al 2010):

Stage	Methodological Focus	Tools/Outputs
1. Exploration	Literature review and precedent analysis	Annotated research database; conceptual sketches
2. Ideation	CAD modelling and iterative prototyping	SolidWorks modelling, exploded diagrams, 3D print assembly testing
3. Prototyping	Fabrication of aluminium wall mock-ups	CNC machining, T-slot extrusion assembly, gasket testing, prototype assembly
4. Evaluation	Comparative analysis and reflective documentation	Photo logs, performance mapping, field notes

This cyclical process allowed reflection-in-action, where material and functional decisions were continuously informed by testing and observation (Schön 1983).

3.6 Design Iteration Log

Each design update of the Proteus wall evolved through prototyping and literature synthesis. The following table summarises the progression.

Table 3.2 Design Iteration Log

Update	Design Focus	Change Implemented
1	Research, initial idea for upgradable house	
2	Modular Design	Design to be modular
3	Frame Design	Removable frame and wall testing, 1:25 scale model
4	Cladding	Decision to make use of aluminium wall cladding
5	Frame Geometry	Shift from steel to 50×50 mm aluminium T-slot
6	Wall Panel	Make each wall panel its own assembly/part
7	Internal Cavity	Added 100 mm service void for wiring/plumbing/insulation
8	Fastening	Developed insertable M8 T-slot bolt with anti-twist design
9	Weatherproofing	Added 1 mm rubber compression seal between modules
10	Removable internal cladding	Make use of removable cladding fasteners
11	External cladding	Weld slats with thread to make external frame semi-permanent
12	Further Prototype Modelling	Fabricated 1:12 magnetic prototype
13	Future Adaptation	Added door/window variants using future custom aluminium extrusions
14	1:1 Prototype	Determined fasteners, tested construction/installation

3.6 Methods of Data Collection

3.6.1 Literature and Document Review

To compile theoretical and empirical insights from sources relevant over 2005-2025, a comprehensive literature review was undertaken that stretched across databases such as Scopus, ScienceDirect, Google Scholar, etc. Supplementary materials included policy documents (e.g. ABCB 2022; GBCA 2024) and Australian housing market analyses (NHFIC 2023; OECD 2021). They underpinned design decisions for sustainability metrics, modular standards, life-cycle considerations.

3.6.2 Field Observations

Field observation visits to the project site of Wild Modular brought first hand insights into on-site assembly, precision control and the sequence of Assembly. Notes and photographs document frame composition, module junctions, material selection and finishing techniques. Thermal bridging at steel interfaces and difficulty accessing service lines post-installation are highlighted by observations. These are the main reasons Proteus has an accessible internal cavity.

3.6.3 Design Prototyping

Digital modelling was undertaken in SolidWorks to generate precise parametric assemblies of the 50 × 50 mm T-slot aluminium frame. Prototypes were then physically fabricated using CNC-machined aluminium extrusions, Fast Mount clips, M8 insertable T-slot bolts, and 3 mm aluminium cladding panels. Tolerance studies and assembly trials validated the feasibility of insertable bolt design and compression sealing performance.

The prototypes were visually and functionally documented using photographic and dimensional analysis. No destructive or mechanical stress testing was performed due to scope constraints, but findings were qualitatively compared with similar systems documented in (Lawson et al 2014) and (Ferdous et al 2021).

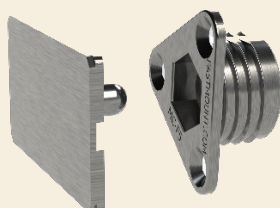


Figure 3.3: Fast mount Clip renders

3.7 Data Analysis and Evaluation Criteria

Data analysis combined *comparative evaluation* and *reflective synthesis*. Observations and prototype performance were evaluated against the following criteria, derived from DfMA/DfD and NCC Class 1 standards:

Criterion	Description	Source
Manufacturability	Ease of fabrication, tolerance consistency, and modular standardisation	(Arashpour et al 2023)
Assembly efficiency	Reduction of fixings and tools required for installation	(Fard et al 2017)
Disassembly and repairability	Accessibility for maintenance and reconfiguration	(Crowther 2021); (Durmisevic 2019)
Material sustainability	Recyclability and embodied energy profile	(Schweizer et al 2022); (AAC 2022)
Regulatory compliance	Alignment with NCC Class 1 provisions (fire, structure, energy)	(ABCB 2022)
User adaptability	Capacity for interior/exterior reconfiguration and upgrades	(Schneider & Till 2007)

These criteria provided an evidence-based structure for assessing design outcomes and situating them within broader architectural discourse.

3.8 Ethical Considerations

The project complied with the Western Sydney University Human Research Ethics guidelines, including informed consent for Interviews. Site documentation at Wild Modular was conducted with permission, ensuring no proprietary information or identifiable personnel were recorded. The research involved no human participants in testing phases and posed minimal ethical risk.

3.9 Limitations

Several limitations shaped the study's scope:

1. Empirical testing constraints – mechanical and weatherproofing performance was not laboratory-tested due to resource limitations.
2. Generalisability – findings are context-specific to Australian modular systems and may not fully translate internationally.
3. Economic assessment – while lifecycle costs were discussed qualitatively, no full cost–benefit model was applied.
4. Prototype scale – the research focused on a single 1:1 wall section, not a full module or building assembly.

Despite these limitations, the study establishes a methodological framework for future quantitative testing and standardisation of adaptive modular wall systems.



Figure 3.4: Wild Modular site observation photograph

Chapter 4: Industrial Design Applied Research and Prototype Development

4.1 Overview

This chapter documents the applied design and fabrication processes undertaken to develop the Proteus adaptive wall system from conceptual sketches through to a full-scale prototype. The research follows the DfMA and DfD principles established in Chapter 2, with iterative prototyping used to translate theoretical criteria into manufacturable and testable components. The development process integrates digital modelling, 3D printing, physical assembly, and material testing to evaluate mechanical performance, tolerance control, and modular alignment within the limits of an Honours-level industrial design project.

No destructive or environmental testing was conducted. Instead, performance expectations are determined from comparable aluminium façade research (Lawson et al 2014; Bai et al 2020). The system is therefore described as *designed to meet the intent of NCC 2022 Class 1* and *AS/NZS 1170* provisions.

Future testing should include R-value verification under *AS/NZS 4859.1:2018*, acoustic measurement via *AS/NZS ISO 717.1:2004*, and fire testing under *AS 1530.1* and *AS 1530.3*. All claims regarding weather and structural performance are currently qualitative and literature-based.

4.2 Digital Modelling and Parametric Framework

Initial design development was completed in SolidWorks 2024, enabling the creation of parametric assemblies reflecting the 50 × 50 mm European standard aluminium extrusion (EN 6063-T5). Each component—corner bracket, panel fastener, and removable cladding interface—was modelled with precise tolerance stacks to ensure manufacturability within ± 0.5 mm.

Digital simulation allowed the evaluation of component compatibility, extrusion slot alignment, and stress distribution under nominal residential wall loads, referencing analytical methods described by Lawson, Ogden & Goodier (2014) and Ferdous et al (2019). Although finite-element analysis (FEA) was not the primary focus, sectional studies were conducted to visualise potential stress points around through-bolt interfaces.

CAD data were subsequently exported as DXF files for CNC machining and STL files for additive manufacturing of scaled prototypes, establishing a direct digital-to-physical workflow consistent with DfMA principles (BCA 2019; Arashpour et al 2023).

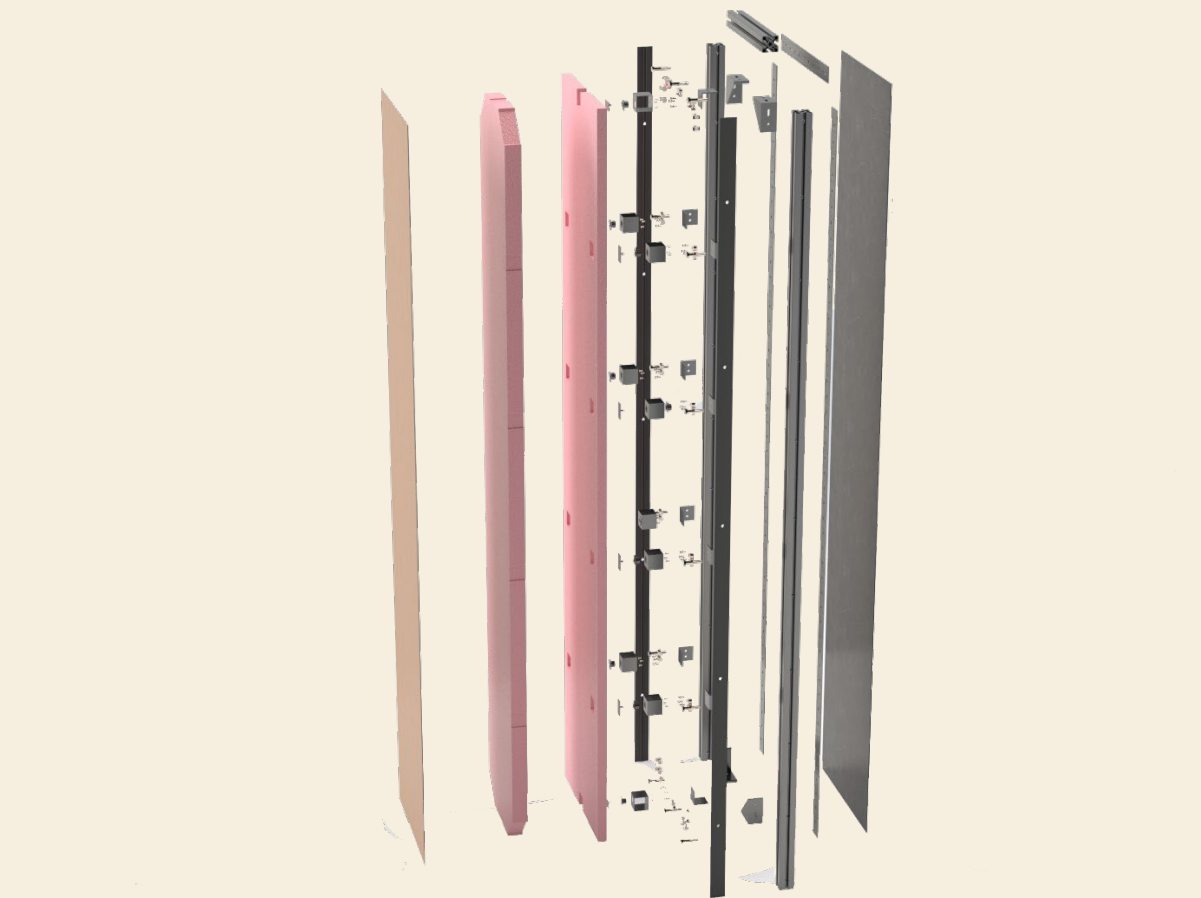


Figure 4.1: Parametric SolidWorks assembly of the adaptive wall frame

4.3 Prototyping and 3D Printing for Validation

Additive manufacturing was employed as a low-cost validation tool prior to full-scale fabrication. 3D-printed components were produced in PLA at 0.2 mm layer resolution to replicate key structural joints, including:

- the insertable M8 T-slot bolt,
- corner bracket assemblies, and
- wall-to-frame interfaces.

This stage enabled physical evaluation of assembly logic, rotational constraints, and clearances before committing to aluminium machining. Rapid prototyping is widely recognised as an effective means of reducing design iteration time and material waste (Berman 2012; Ford & Despeisse 2016). (Ngo et al 2018) further note that early 3D-printed verification of component fit supports precision manufacturing and reduces cost overrun during tooling.

The 3D-printed extrusions and brackets confirmed that insertable bolts could be guided into the T-slot without end-feeding while maintaining axial stability—a critical validation for field assembly where continuous slot access is not feasible.

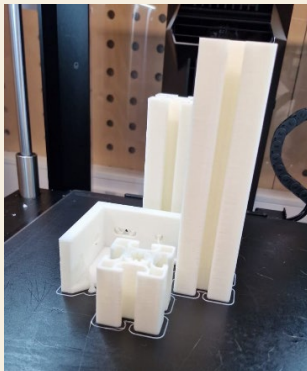


Figure 4.2: 3D-printed extrusion and corner bracket prototypes



Additionally a 1:12 house prototype demonstrated full interchangeability between wall, door, and window panels, using magnetic joints to simulate fastener engagement and reconfiguration. This small-scale study was supported by finish samples of various aluminium coatings, providing visual and tactile feedback on future product iterations.

(Figure 4.3: 1:12 Magnetic Model Demonstrating Modular Interchangeability)

4.4 Full-Scale Fabrication and Material Assembly

Following digital and additive verification, full-scale prototyping was conducted using 50-series aluminium extrusions cut to 500 mm and 1 000 mm lengths. Corner joints were assembled using machined aluminium brackets fixed by M8 bolts and nyloc nuts. Internal and external panels were fabricated from 3 mm aluminium sheet Textured wall paper to emulate both interior wood-grain and exterior marble-like textures

The assembly incorporated two cavity zones:

- a 100 mm insulated service cavity, accommodating standard residential insulation batts; and
- a 50 mm wiring and plumbing cavity, positioned behind removable interior panels.

A 1 mm EPDM rubber compression seal was inserted between panel edges to achieve weatherproofing through compression rather than sealant adhesion, aligning with AS 4284 Façade Weathertightness testing methodology.

Through-bolts (M10 × 125 mm) mechanically connected adjacent wall frames, maintaining alignment and rigidity while allowing disassembly. This bolted strategy follows reversible-design recommendations by Durmisevic (2019) and Crowther (2022), who emphasise the value of mechanical fastening for circular material recovery.

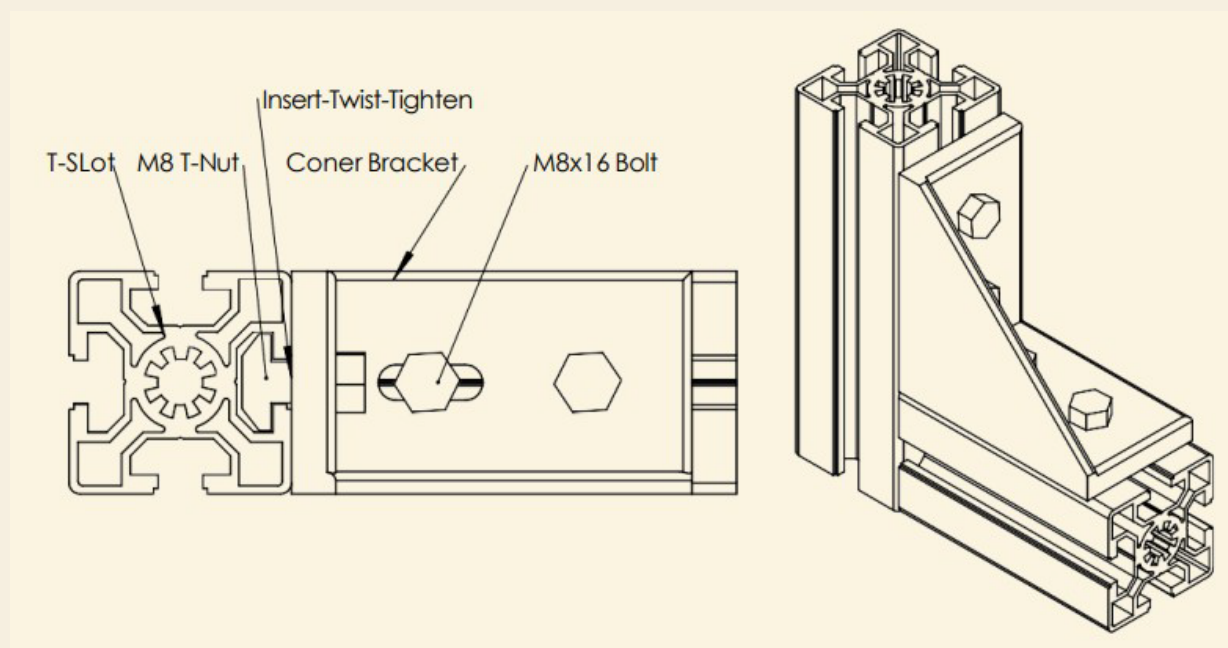


Figure 4.3: Insertable M8 T-slot bolt connection detail/Corner bracket assembly

4.5 Testing and Evaluation

Testing focused on qualitative assessment of assembly efficiency, fit tolerance, weatherproofing compression, and user ergonomics. The prototype was assembled and disassembled multiple times to evaluate fastener accessibility and gasket recovery. Observations confirmed that all panels could be removed using a spanner, screwdriver and moderate manual force, validating the DfD criterion of tool-light disassembly (Crowther 2022).

During assembly, the compressive seal achieved continuous contact across vertical and horizontal joints. While quantitative weathertightness testing per AS 4284 was outside project scope, visual inspection showed no daylight penetration, suggesting effective preliminary sealing. Future research should perform controlled hydrostatic and air-pressure testing to validate this performance empirically.

Structural load capacity, sound insulation, and thermal resistance were not formally tested; however, reference modelling based on (Lawson et al 2014) and (Schweizer et al 2022) indicates that aluminium wall assemblies of similar thickness achieve adequate stiffness for partitions. These aspects remain identified for postgraduate or industry-funded continuation.



Figure 4.4: Full-scale prototype assembly

4.6 Design Iteration and Refinement

Several key refinements emerged through iterative testing:

1. Fastener ergonomics: Early insertable bolt designs were prone to rotational slip during tightening. The revised M8 design incorporated a sharp edged locking shape to prevent rotation whilst tightening, allowing for ease of assembly.
2. Panel alignment: Minor bowing was addressed by increasing extrusion rigidity through doubled top rails, echoing DfMA tolerance strategies noted by (Gibb & Isack 2003).
3. Bracket stability: 3D-printed corner connectors were replaced with solid aluminium components to enhance rigidity and allow repeatable welding.
4. Ease of access: Interior panels were modified to accept *FastMount clips*, enabling tool-light removal for service access while maintaining secure engagement during use.
5. The exterior panels were welded with slats which house threaded inserts, this allows for appropriate force application, which allows for adequate compression of the rubber seal. This fact also allows for the external panels to be more appropriately fixed for external conditions.

These refinements emphasize the reflective, iterative nature of design-led research as advocated by (Schön 1983) and (Niedderer, Cain & Walters 2023)—in which making and testing function like twin engines of invention

4.7 Alignment with DfMA and DfD Principles

The prototype successfully embodies several DfMA and DfD criteria outlined in (BCA 2019) and (Ellen MacArthur Foundation 2019) frameworks:

- DfMA: modular standardisation (50 mm grid), minimal part count, and pre-machined extrusion profiles reduce on-site operations and human error (Arashpour et al 2023).
- DfD: all fixings are mechanical, reversible, and accessible; cladding and insulation layers can be removed without structural disruption (Durmisevic 2019).
- Circularity: aluminium and EPDM rubber allow for near-complete recyclability consistent with Australian Aluminium Council (2022) data on closed-loop material recovery.

These alignments validate the Proteus system's compliance with contemporary sustainability metrics and reinforce its potential role in future adaptive housing models.

4.8 Future Testing and Development

Although the prototype achieved its intended design objectives, further empirical testing is recommended:

1. Structural certification in accordance with AS/NZS 1170 Structural Actions.
2. Thermal and acoustic testing to establish R-values and transmission loss coefficients.
3. Accelerated weathering tests to verify seal longevity and corrosion resistance under Australian climatic conditions.
4. Lifecycle assessment (LCA) comparing aluminium to timber or steel equivalents (Kamali & Hewage 2016; Aye et al 2020).

Additionally, digital integration with *Building Information Modelling (BIM)* platforms could facilitate parametric customisation and supply-chain optimisation (Salman, Khalfan & Maqsood 2012; Sacks et al 2018).

Future development will include **custom extrusions for double-glazed window and door frames**, enhancing performance and expanding functional typologies. The design's potential scalability to vertical construction—such as high-rise façades—will be investigated following engineering certification to *AS/NZS 1170*. These applications align with DfMA and DfD principles, supporting efficient assembly, replacement, and reusability at scale (Azhar et al 2012; Lawson et al 2014).

4.9 Summary

The design and fabrication of the Proteus adaptive wall system demonstrate how industrial design methods—grounded in DfMA and DfD—can produce an innovative, serviceable building component capable of integration within modular housing systems. The combined use of digital modelling, additive validation, and full-scale aluminium prototyping exemplifies design-led research that is both materially grounded and conceptually rigorous.

The chapter concludes that a hybrid workflow of CAD-CNC-3D print-assembly provides a scalable framework for design testing in Australian modular-construction contexts, enabling future cross-disciplinary collaboration between industrial designers, engineers, and manufacturers.

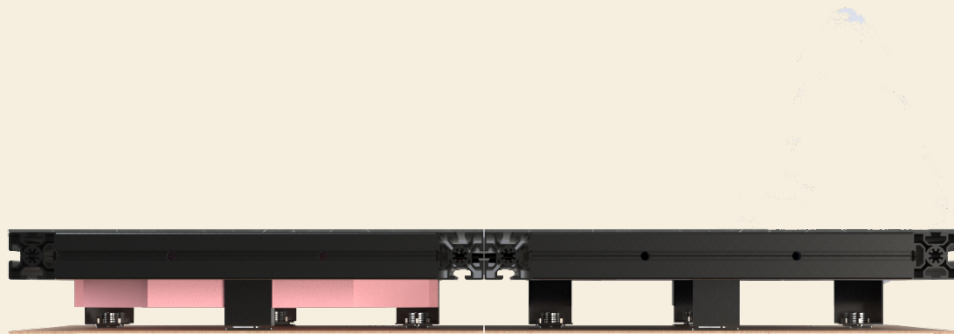


Figure 4.5: View of wall assembly showing cavities and insulation and service cavities

Chapter 5: Findings and Evaluation

5.1 Overview

This chapter evaluates the Proteus system's adaptive wall--considering the research objectives, design standards, and building codes discussed in previous chapters. Design testing, site observations and comparing benchmarks with other related codes and actualities. The assessment focuses upon three integrated measures of performance:

1. Technical and functional performance,
2. Environmental and circular-economy impact, and
3. Regulatory and compliance alignment under the National Construction Code (NCC 2022) for Class 1 housing.

Findings demonstrate that the wall system effectively integrates Design for Manufacture and Assembly (DfMA) and Design for Disassembly (DfD) principles within a practical, buildable framework that advances current modular housing practice in Australia.

5.2 Technical and Functional Evaluation

The Proteus wall prototype achieved a modular structural configuration capable of consistent alignment, disassembly, and reassembly across multiple test cycles. The frame maintained its geometric stability due to the inherent stiffness of aluminium extrusions, corner brackets and the use of through-bolted joints along with custom designed T-Slot bolts.

Unlike traditional modular systems, this design enables replacement of individual panels or structural members without disturbing adjacent assemblies, with the additional ability of easy extension. This functional outcome aligns with DfD performance criteria established by (Crowther 2022) and (Durmisevic 2019), who both identify reversibility and accessibility as central to material longevity and adaptability.

The system's insertable T-slot bolts provided tool-light assembly, confirming ergonomic ease and compliance with DfMA simplification strategies (BCA 2019; Arashpour et al 2023). The rubber compression gasket achieved effective surface continuity, supporting NCC Part 3.5.3 requirements for external moisture exclusion (ABCB 2022).



Figure 5.1: Compression seal

Field observation and assembly documentation further confirmed:

- Panels could be removed without mechanical damage, ensuring serviceability;
- The modular cavity system accommodated insulation and wiring separation;
- Corner brackets maintained joint rigidity during repeated installation cycles.

Collectively, these findings support the proposition that a design-led, demountable wall system can achieve technical feasibility while enhancing modular housing adaptability.

5.3 Environmental and Circular Performance

The environmental outcomes of the system are assessed qualitatively using life-cycle indicators derived from (Aye et al 2020) and (Schweizer et al 2022).

Because aluminium is infinitely recyclable, the choice of material supports circular-economy objectives under the Green Star Buildings v2 framework (GBCA 2024). The closed-loop recyclability of extruded aluminium reduces embodied energy by up to 95% compared with primary production (Australian Aluminium Council 2022).

The Proteus system's mono-material construction (extrusions, brackets, and panels all aluminium) simplifies end-of-life separation and minimises contamination, addressing key waste-reduction objectives under the National Waste Policy Action Plan 2023–2030 (DCCEEW 2023). The inclusion of reversible mechanical joints ensures that disassembly, not demolition, defines the end-of-life phase—transforming walls from static fixtures into recoverable assets, which is further emphasised through the use of screw pier foundations in place of concrete slabs.

(Crowther 2022) suggests that component-level design decisions such as fastener selection have cumulative impacts on national waste streams. By eliminating adhesive bonds and replacing them with bolts and compression seals, the Proteus design aligns with the material stewardship ethos promoted by (UNEP 2022) and (Ellen MacArthur Foundation 2019).

While quantitative lifecycle assessment was beyond project scope, the comparative analysis below (Table 5.1) summarises the sustainability potential relative to conventional framed wall systems.

Table 5.1 Comparative Summary of Modular Wall System Performance

Performance Dimension	Conventional Wall (Timber/Steel Stud)	Proteus Wall System (Aluminium Modular)		Supporting References
Disassembly	Destructive (nailed/screwed joints)	Reversible (mechanical bolts & clips)		Crowther 2022; Durmisevic 2019
Recyclability	Low–moderate	High, Use of aluminium		AAC 2022; Schweizer et al 2022
Maintenance	Difficult access	Full access via removable panels		Lawson et al 2014
Embodied energy	Moderate–high	Moderate (offset by recyclability)		Aye et al 2020
Waste at end-of-life	20–30% material loss	Minimal, Main frame and cladding may be re-used/recycled		Kamali & Hewage 2016
Adaptability	Minimal	High (service and finish interchangeability)		Schneider & Till 2007
		Modular (Typical)	Proteus Design (Anticipated)	
Construction Time	9–12 months	4–6 months	Designed for rapid assembly within factory conditions	Ferdous et al 2019
Material Waste	20–30%	5–10%	Anticipated <10% through off-site precision cutting	Davis et al 2019
Site Impact	High (earthworks, concrete)	Moderate	Minimal – screw-pier foundations; no concrete slab, can be removed	Bai et al 2020
Reusability	Low	Moderate	High – mechanical fixings and mono-material system	Kamali & Hewage 2016
Embodied Carbon	~500–700 kg CO ₂ /m ²	20–30% lower	Expected reduction pending testing	Bai et al 2020

5.4 Alignment with Regulatory Frameworks (NCC 2022)

The Proteus system was evaluated against key performance provisions of NCC 2022, Vol. 2 (Class 1 buildings). Table 5.2 summarises compliance alignment and potential considerations for certification.

Table 5.2 Clause-by-Clause Summary of NCC 2022 Compliance Alignment

NCC Section	Relevant Clause	Proteus System Response	Notes
Structure	3.0.4; AS/NZS 1170	Aluminium framing meets non-loadbearing wall requirements; through-bolts ensure lateral stability.	Engineering certification required for commercial use.
Fire Safety	3.7.1	Testing needed, Fire rated Aluminium cladding panels are currently in-market.	Fire-stopping details to be developed for multi-module junctions.
Weatherproofing	3.5.3; AS 4284	1 mm rubber compression seal prevents water ingress through joints.	Full AS 4284 testing recommended.
Energy Efficiency (Section J)	J1.2–J3.3	100 mm insulated cavity supports R2.5 equivalent performance.	Thermal bridging assessment pending.
Condensation Management	3.8.7	Airtight compression joints and insulation barrier meet condensation control principles.	Hygrothermal simulation required.
Sound Insulation	3.8.6	Not tested; potential for acoustic infill.	Future testing required.
Openings & Fixtures	AS 2047	Window/door modules to integrate via compatible aluminium framing.	Further detailing needed.

5.5 Comparative Industry Benchmarking

Benchmarking the Proteus system against existing Australian modular housing solutions (e.g. Wild Modular, Modscape, and ArchiFrame) reveals several distinctions. While most industry models rely on welded steel frames with fixed internal linings, the Proteus wall introduces replaceable and serviceable layers, aligning with DfD literature (Durmisevic 2019). This layered configuration—outer cladding, structural frame, insulation cavity, service cavity, and removable interior panel—echoes the reversible “peel-back” assembly proposed by (Ferdous et al 2019) for modular high-rise façades.

In contrast to conventional steel modules, the Proteus assembly achieves *functional modularity* (component-level reconfiguration) rather than only *volumetric modularity* (room-level prefabrication). As (Kamali and Hewage 2016) argue, system modularity rather than unit modularity determines long-term circular performance in housing systems.

5.6 User and Industry Feedback

Although no formal user testing was conducted, informal consultations with professionals at Wild Modular provided valuable feedback.

Key reflections included:

- *Positive response* to reversible fixings that allow maintenance access without damaging finishes;
- *Interest in scalability*, particularly for façade retrofits and transportable units;
- *Concern regarding certification pathways*, noting that novel systems require comprehensive documentation for Building Surveyor acceptance.
- *Interest in façade systems for high rise construction.*

These insights reflect findings by (Manuel, Hwang and Shan 2021), who identify regulatory uncertainty as a primary barrier to modular innovation in Australia.



Figure 5.2: Prototype demonstration for industry feedback (SXSW 2025 Tech and Innovation Showcase)

5.7 Limitations of Findings

While results indicate strong qualitative performance, several limitations remain:

- No mechanical testing was performed to quantify shear or bending strength.
- Weather, acoustic, and insulation properties remain to be validated.
- Cost modelling has yet to be performed to assess market competitiveness.
- Engineering certification is pending for aluminium load-bearing capacity.

Nevertheless, the prototype meets its design objectives as a research demonstrator and establishes a robust baseline for future empirical study.

5.8 Summary

The Proteus adaptive wall system demonstrates clear progress toward integrating DfMA and DfD within Australian modular housing.

Key findings confirm that:

- Aluminium-based systems can achieve disassembly, recyclability, repairability, upgradability consistent with circular-economy frameworks (Crowther 2022; Schweizer et al 2022).
- Design-led prototyping validates the manufacturability and ergonomics of adaptive wall systems.
- The system aligns with the structural, thermal, and weatherproofing intents of NCC 2022, pending formal certification.

These results position the Proteus project as an exemplar of how *industrial design practice* can bridge conceptual research and applied construction innovation, establishing a foundation for broader industry adoption of adaptive modular systems in Australia.

Chapter 6: Analysis and Discussion

6.1 Overview

This chapter reflects on the research findings within the theoretical and industrial context of sustainable modular housing. It combines insights from the literature, design testing, and field observations to explain how the Proteus adaptive wall system contributes to and aligns to *Design for Manufacture and Assembly (DfMA)*, *Design for Disassembly (DfD)*, and the *circular economy*. It further places the design within the context Australia's housing policy and sustainability frameworks, analysing how design-led research can influence innovation in construction.

6.2 The Role of Industrial Design in Construction Innovation

Industrial design traditionally focuses on consumer products and manufacturing efficiency. However, this research demonstrates how its principles—precision, modularity, ergonomics, and lifecycle thinking—can be directly applied to architectural systems. (Godin and Beveridge 2021) describe design research as a means of “making knowledge tangible,” and this approach proved vital to bridging conceptual and technical aspects of construction.

In the case of the Proteus system, the iterative design process allowed for *formative testing* at multiple scales: from 3D-printed prototypes validating mechanical logic to full-scale fabrication demonstrating manufacturability. This aligns with the “research-through-design” paradigm explored by (Zimmerman, Forlizzi and Evenson 2010), where each product functions as an evolving idea. The project shows how industrial design can operate as a *knowledge intermediary* between the ambitions of sustainability and the concrete realities of material production.

This positioning aligns with (Ferdous et al 2021), who notes that the success of modular innovation depends on integration between engineering precision and design intuition. Through controlled prototyping, the Proteus research achieves precisely that.

6.3 Application of DfMA and DfD Frameworks

The Proteus adaptive wall system offers a tangible case study in combining DfMA and DfD within one component. DfMA principles guided the reduction of parts, standardisation of dimensions, and the simplification of assembly, consistent with (BCA 2019) and (Gibb and Isack 2003) recommendations for preassembly efficiency. Each element—the T-slot frame, removable panel, and gasketed seal—was designed for reproducibility and tolerance control, allowing for the real possibility of total automation.

Conversely, DfD principles directed attention to reversibility and lifecycle extension. The removable panels, accessible fasteners, and compression-based weatherproofing exemplify design-for-disassembly practices described by (Durmisevic 2019) and (Crowther 2022). By minimising permanent bonds and specifying mono-material aluminium construction, the system aligns with circular-construction frameworks such as those developed by the (Ellen MacArthur Foundation 2019) and (UNEP 2022).

The combining of DfMA and DfD within one system is a notable contribution, as many modular precedents prioritise one principle over the other—efficiency or adaptability—but rarely both. As (Kamali and Hewage 2016) explain, the challenge in sustainable prefabrication lies in factory precision with the allowance for flexibility over time. The Proteus prototype begins to resolve that through component-level innovation.

6.4 Circular Design and Lifecycle Thinking

Circular design reframes buildings as material banks rather than consumable assets. The Proteus system embodies this by allowing panels, frames, and fixings to be retrieved, repaired, and reintroduced into production. Its design supports closed-loop material cycles, consistent with the National Waste Policy Action Plan 2023–2030 (DCCEEW 2023) and Green Star Buildings v2 (GBCA 2024).

While aluminium carries higher initial embodied energy, its infinite recyclability offsets long-term environmental impact (Schweizer et al 2022). (Aye et al 2020) found that modular aluminium systems outperform timber and steel in reuse potential and corrosion resistance, especially in humid climates. Consequently, the Proteus approach supports Australia's broader decarbonisation agenda outlined in the (DCCEEW 2024) Net Zero Plan.

At a conceptual level, the design also addresses issues of housing obsolescence. (Schneider and Hill 2007) argue that housing should adapt as occupants' needs evolve. The ability to replace an interior panel with a door, window, add or remove rooms, with the possibility of adding levels has precisely this flexibility. Thus, the wall becomes an active infrastructure component capable of responding to demographic, technological, and environmental change.

6.5 Regulatory and Industry Implications

From a regulatory standpoint, the system aligns with NCC 2022 and AS/NZS 1170 intent but challenges conventional compliance pathways. Australian construction regulation remains oriented toward large, permanent assemblies (Van Gulijk, Chynoweth & Higgins 2012). Consequently, innovative modular components often face certification barriers not because of performance deficiencies but due to policy misunderstanding.

The Proteus project underscores the need for *performance-based approval frameworks* that accommodate modular implementation. This aligns with insights from (PrefabAUS 2022) and (Infrastructure Australia 2023), which both highlight the need for changes in policy to assist MMC innovation and implementation.

Industrial engagement revealed growing receptivity among modular builders such as Wild Modular, who recognised the system's serviceability advantages but expressed concern about testing costs and certification requirements. These findings mirror those of (Manuel, Hwang and Shan 2021), who identified policy uncertainty as a key barrier to innovation in Australia's prefabrication sector. Addressing these constraints will be essential to moving from prototype to market implementation.

6.6 Human-Centred and Aesthetic Dimensions

Beyond technical performance, the Proteus design responds to human-centred design considerations central to the industrial design practice. The removable interior panels improve accessibility for maintenance and upgrades, reducing the difficulty, time and cost of repair and upgrades. This enhances the users control of their product and aligns with the “right to repair” sentiment increasingly discussed in sustainable design policy (OECD 2021).

Aesthetically, the dual-panel finish—with the user being able to select one of many already available finishes—demonstrates that sustainability need not compromise visual quality. By designing the wall as both a functional and aesthetically pleasing product, the design reinforces the notion that adaptive architecture must also remain desirable. (Leach, Niemann and Steele 2021) highlight that environmental adaptation is as much a cultural as a technical pursuit; design aesthetics play a role in user acceptance which is key market uptake.

6.7 Integration with Digital Fabrication and BIM

Digital fabrication played a central role in the development of the Proteus prototype, illustrating how computer-aided design and additive manufacturing can bridge design and physical realisation. As (Salman, Khalfan and Maqsood 2012) and (Sacks et al 2018) explain, Building Information Modelling (BIM) and digital prototyping allow for parametric precision, real-time simulation, and data-rich documentation that support both DfMA efficiency and long-term asset management.

The project’s workflow—from SolidWorks modelling to 3D-printed mock-ups and CNC-machined parts—reflects an emerging hybrid model of *distributed digital fabrication*. Such workflows are increasingly essential to the Australian context, where prefabrication must adapt to regional manufacturing capacities (Arashpour et al 2023).

We can see this being interested in new buildings such as the advanced manufacturing facility located in Bradfield.

Integration with BIM would enable automated generation of cut lists, tolerance checks, and assembly guides, further supporting large-scale implementation.

6.8 Education and Research Contribution

At the academic level, the project demonstrates how *Research-through-Design (RtD)* can operate as a legitimate mode of testing and is key in industrial design education. The iterative, evidence-based process used to create and evaluate the Proteus prototype produced not only a design but also a transferable research framework for future research. (Niedderer, Cain and Walters 2023) and (Godin and Beveridge 2021) emphasise that RtD contributes to disciplinary knowledge by producing *exemplars*—designs that embody new methods or principles.

By articulating its own decision-making process, the Proteus project contributes a methodological precedent for integrating sustainability metrics and compliance logic into design-led research. It also illustrates the value of university–industry collaboration: practical engagement with Wild Modular contextualised the theoretical work and allowed exposure to real-world manufacturing constraints, which may not have been possible without this scale and type of project.

6.9 Broader Industry Impact

The implications of this research extend beyond the prototype. As modular and prefabricated construction gains momentum under Australia’s Housing Strategy 2041 (NSW Government 2022), systems like Proteus could support scalable, circular, and user-adaptive housing solutions.

Its modularity enables diverse applications—from temporary emergency housing to permanent dwellings—with the ability to upgrade rather than replace. Such a philosophy supports the shift from “linear” to “adaptive” housing stock envisioned by (Steinhardt and Manley 2016) and (Zhai, Reed and Mills 2014).

By embedding disassembly and reuse into the product DNA, Proteus advances the cultural shift toward designing for the future—a key step in meeting the UN Sustainable Development Goals (UN 2015) and Australia’s net-zero transition (DCCEEW 2023).

6.10 Limitations and Future Research Directions

While the research produced useful insights, it is halted by the lack of quantitative performance data. As highlighted in Chapter 5, no structural, acoustic, or thermal testing was undertaken. Future studies should involve full-scale laboratory assessment under AS/NZS 1170 and AS 4284 conditions.

Economic feasibility also requires further study. Although aluminium offers recyclability advantages, its upfront cost may hinder adoption unless offset by lifecycle savings. Cost–benefit modelling like that used by (Jiang, Wu and Qian 2018) could clarify financial viability.

To remove another barrier, engaging policymakers and certification bodies early would support mainstream adoption of DfD-enabled modular systems.

Finally, expanding digital integration with BIM and exploring open-source manufacturing models could facilitate an easier way for users to design their own product, through live updates in assembly.

6.11 Summary

The Proteus adaptive wall system represents a synthesis of theory and practice that advances Australia’s pursuit of sustainable, circular housing innovation. It demonstrates that industrial design-led research can contribute materially to the rethinking of construction systems, merging digital design, sustainable materials, and modular assembly.

Through the deliberate application of DfMA and DfD principles, the project transforms the wall from a static solution into a dynamic infrastructure for adaptation, maintenance, and reuse. The work thus aligns with industrial design’s vital role in reimagining the built environment—technically, environmentally, and culturally.

Chapter 7: Conclusion and Significance

7.1 Summary of Aims and Findings

Proteus represents an adaptive wall system that redefines modular housing as a scalable, circular, and affordable solution to traditional construction. The system benefits from design-for-manufacture efficiency and the flexibility it offers to occupants whose needs change. New house owners have the ability to begin with a compact dwelling that can be expanded later by interconnecting modules with foundations that are reusable and reversible, in addition to being environmentally friendly.

The system is constructed of lightweight aluminium with service cavities that provide ease of transport, assembly, and maintenance and is supported by a library of standardised components for commercial scalability. Finally, the fact that Proteus has the potential to be extended to high-rise façade systems demonstrates its cross-sector utility within the expanding Australian modular-construction market.

Through a research-through-design (RtD) methodology, the project combined theoretical design and analysis with hands-on experimentation and prototyping, culminating in the fabrication of a full-scale prototype. The system's design incorporated Design for Manufacture and Assembly (DfMA) and Design for Disassembly (DfD) principles, resulting in a modular framework composed of aluminium extrusions, mechanical fasteners, removable panels, and gasketed seals.

Findings confirmed that:

- The Proteus system achieved functional modularity and disassembly, allowing for panel replacement and maintenance access without destructive intervention.
- The design theoretically met the intent of NCC 2022 provisions for moisture, insulation, and durability, with potential for further compliance certification.
- Material and design choices aligned with circular-economy principles and national sustainability frameworks such as Green Star Buildings v2 and the National Waste Policy Action Plan (2023–2030).
- Engagement with industry partners demonstrated the system's potential applicability and identified certification as a key adoption challenge.

Collectively, these results show that industrial design-led innovation can contribute to advancing modern methods of construction (MMC) in Australia through a focus on precision, modularity, and lifecycle thinking.

7.2 Theoretical and Industrial Contributions

The research contributes to three fields:

1. Industrial Design Methodology – It validates *design as a method of inquiry*, demonstrating how iterative making and testing can generate new forms of construction knowledge. The project aligns with (Niedderer, Cain and Walters 2023) in framing design prototyping as both process and outcome.
2. Circular and Adaptive Architecture – The Proteus system contributes a built exemplar of DfD applied at the component level. It extends the conceptual framework of (Durmisevic 2019) and Crowther 2022) by demonstrating disassembly through reversible mechanical joints rather than bonded systems.
3. Australian Modular Housing Practice – The research addresses the innovation gap identified by (Manuel, Hwang and Shan 2021), showing that design-led experimentation can inform next-generation modular solutions compatible with Australian standards and climate.

The outcome therefore bridges the conceptual, technical, and policy domains of sustainable construction, demonstrating that design research can serve as a mechanism for both innovation and policy influence.

7.3 Limitations

While successful in demonstrating technical feasibility, the research was limited in several respects:

- Quantitative testing (structural, thermal, acoustic) was not conducted.
- Engineering certification was outside project scope.
- Cost and lifecycle analyses were indicative rather than empirical.
- Prototype fabrication occurred under controlled studio conditions rather than field environments.

These constraints reflect the scope of an Honours-level study, but they also identify clear pathways for future collaboration with engineering and construction disciplines.

7.4 Future Research and Development

Further advancement of the Proteus adaptive wall system should combine technical validation, environmental assessment, digital integration, and broader prototyping. The following priorities are proposed for future research and industrial collaboration.

1. Engineering Validation

Independent structural and weather-tightness testing should be performed in accordance with *AS/NZS 1170 Structural Design Actions* and *AS 4284 Testing of Building Façades*. Load testing across wall spans, joint behaviour under wind pressures, and façade water-penetration resistance must be verified by accredited laboratories to demonstrate compliance with *NCC 2022 Volume 2 Class 1 requirements* (ABCB 2022). These results will provide formal engineering certification and inform future connection detailing.

2. Lifecycle Analysis

A comprehensive life-cycle assessment (LCA) should measure the embodied carbon, operational energy, and recyclability of the aluminium-based system compared with equivalent steel- and timber-framed walls. Using recognised frameworks (Kamali & Hewage 2016; Bai et al 2020) will clarify the Proteus system's overall environmental performance and identify opportunities for carbon reduction through material substitution and process optimisation.

3. Digital Integration

Integration with Building Information Modelling (BIM) should automate component documentation, tolerance control, and logistics tracking. Linking the Proteus wall design to parametric DfMA workflows can streamline fabrication sequencing and error checking (Azhar et al 2012; Lawson et al 2014). This digital workflow would also allow manufacturers to generate live assembly drawings and inventory schedules directly from the model, improving constructability and transparency. Whilst noting the importance of the ability for the user to design for themselves, with easy transparent into a manufacturing setting

4. Commercialisation Pathways

Partnerships with modular-housing manufacturers, certifiers, and state regulators are essential to translate the prototype into a market-ready system. Collaboration would support alignment with the *NCC 2022 Deemed-to-Satisfy* pathway, clarify certification procedures, and ensure early-stage product accreditation (Ferdous et al 2019; Davis et al 2019). Engagement with Wild Modular and similar organisations could accelerate adoption and standardisation across Australian modular projects.

5. Expanded Prototyping

Future prototyping should include window, door, and façade variants utilising the same aluminium T-slot logic. These iterations would test interchangeability, scalability, and service integration for multi-module configurations. Expanded physical mock-ups and 3D-printed sub-assemblies will validate repeatability and inform ergonomic and aesthetic refinements prior to production (Lawson et al 2014).

6. Insulation Testing

Thermal-performance testing should be undertaken to understand and prove the R-value of the wall assembly under *AS/NZS 4859.1:2018 Thermal Insulation Materials for Buildings*. Laboratory measurement using a guarded hot-box or heat-flow-meter apparatus will confirm compliance with *NCC 2022 Section J Energy Efficiency*. Comparative testing of insulation types within the 100 mm service cavity will also evaluate junction losses at corners and openings. Noting the existing design decision of zero, metal to metal contact from exterior to interior

7. Sound Testing

Acoustic testing is required to determine airborne and impact sound-transmission levels following *AS/NZS ISO 717.1:2004 Acoustics — Rating of Sound Insulation in Buildings*. Field and laboratory measurements should establish the weighted sound-reduction index (R_w) for the complete wall system, including joints and removable panels. Results will validate suitability for multi-residential and high-density housing where acoustic separation is essential.

8. Fire Testing

Comprehensive fire-performance testing should verify that aluminium and internal components meet *AS 1530.1* and *AS 1530.3 Methods for Fire Tests on Building Materials*. Testing should examine non-combustibility, heat transfer across cavities, and smoke or toxic-gas emissions. Collaboration with a NATA-accredited laboratory will support certification for external and internal wall assemblies under *NCC 2022 Volume 2 Part 3.7.1 Fire Safety*.

9. High-Rise Façade Integration

Explore adaptation for tall-building façades, assessing vertical load transfer and wind-pressure behaviour under *AS 4284*.

10. Modular Expansion for Housing Affordability

Model incremental-growth scenarios enabling small starter dwellings to expand over time, supporting financial accessibility and reduced embodied carbon.

7.5 Significance of the Research and Conclusion

The Proteus adaptive wall system demonstrates that industrial design can make a tangible contribution to sustainable construction in Australia. It uses design as both a technical and personal practice—one capable of transforming how buildings are designed, built, and adapted.

By integrating digital fabrication and circular design, the project uses and demonstrates a holistic approach to design thinking and professional practice. It also emphasises the importance of industry and governmental collaboration in addressing the housing challenges facing Australia—affordability, adaptability, and sustainability.

In conclusion, the research offers not only a working prototype but a framework for how design-led inquiry can shape the evolution of the built environment towards a more flexible, efficient, and regenerative future.

Reference List

ABCB (Australian Building Codes Board) 2022, *National Construction Code 2022, Volumes One–Three*, Australian Building Codes Board, Canberra.

ABS Australian Bureau of Statistics. (2021, September). *Building Activity, Australia*. ABS. <https://www.abs.gov.au/statistics/industry/building-and-construction/building-activity-australia/sep-2021>.

Arif, M & Egbu, C 2010, 'Making a case for offsite construction in China', *Engineering, Construction and Architectural Management*, vol. 17, no. 6, pp. 536–548, doi:10.1108/09699981011090170.

Salman, A, Khalfan, M & Maqsood, T 2012, 'Building information modelling (BIM): now and beyond', *Australasian Journal of Construction Economics and Building*, vol. 12, no. 4, pp. 15–28, doi:10.5130/ajceb.v12i4.3032.

Bai, Y, Ferdous, W, Manalo, A & Ngo, T 2020, 'New advancements, challenges and opportunities of multi-storey modular buildings – A state-of-the-art review', *Engineering Structures*, vol 183, doi:10.1016/j.engstruct.2019.01.061

Baldwin, A, Poon, C, Shen, L, Austin, S & Wong, I, 2009, Designing out waste in high-rise residential buildings: Analysis of precasting methods and traditional construction volume 34, issue 9, pages 2067-2073, doi:20010.1016/j.renene.2009.02.008.

Blismas, N & Wakefield, R 2009, 'Drivers, constraints and the future of off-site manufacture in Australia', *Construction Innovation*, vol. 9, pp. 72–83, doi:10.1108/14714170910931552.

Kilinc, N, Ozturk, G, Yitmen, I, 2015, 'The Changing Role of the Client in Driving Innovation for Design-build Projects: Stakeholders' Perspective', *Procedia Economics and Finance*, Volume 21, Pages 279-287, doi.org/10.1016/S2212-5671(15)00178-1.

Building and Construction Authority (BCA, Singapore) 2019, *Design for Manufacture and Assembly (DfMA) and Prefabricated Prefinished Volumetric Construction (PPVC) Guidebook*, Building and Construction Authority, Singapore.

Crowther, P 2022, *Exploring the Principles of Design for Disassembly through Design-led Research*, *IOP Conference Series Earth and Environmental Science*, doi:10.1088/17551315/1101/6/062031

Davis, P, Gajendran, T, Vaughan, J & Owi, T 2016, 'Assessing construction innovation: theoretical and practical perspectives', *Construction Economic and Building*, doi:10.5130/AJCEB.v16i3.5178

Ferdous, W, Bai, Y, Ngo, TD, Manalo, A & Mendis, P 2019, 'New advancements, challenges and opportunities of multi-storey modular buildings – a state-of-the-art

review', *Engineering Structures*, vol. 183, pp. 883–893, doi:10.1016/j.engstruct.2019.01.061.

GBCA (Green Building Council of Australia) 2021, *Green Star Buildings: A Guide to Australia's Sustainability Rating System*, GBCA, Sydney.

Kamali, M & Hewage, K 2016, 'Life cycle performance of modular buildings: a critical review', *Renewable and Sustainable Energy Reviews*, vol. 62, pp. 1171–1183, doi:10.1016/j.rser.2016.05.031.

Lawson, M, Ogden, R & Goodier, C 2014, *Design in Modular Construction*, CRC Press, London. doi: 10.1201/b16607

Mao, C, Shen, Q, Pan, W & Ye, K 2013, 'Major barriers to off-site construction: the developer's perspective in China', *Journal of Management in Engineering*, vol. 31, no. 3, 04014043, doi:10.1061/(ASCE)ME.1943-5479.0000246.

Manalo, A, Aravinthan, T, 2012, 'Behaviour of glued fibre composite sandwich structure in flexure: Experiment and Fibre Model Analysis', *Materials and Design*, vol. 112, pp. 478–488, doi: 10.1016/j.matdes.2012.02.052

Rahman, 2014, 'Barriers of implementing modern methods of construction', *Journal of Management in Engineering*, vol. 30, no. 1, pp. 69–77, doi:10.1061/(ASCE)ME.1943-5479.0000173.

Sacks, R, Eastman, C, Lee, G & Teicholz, P 2018, *BIM Handbook: A Guide to Building Information Modeling for Owners, Designers, Engineers, and Contractors*, 3rd edn, John Wiley & Sons, Hoboken.

Steinhardt, DA & Manley, K 2016, 'Adoption of prefabricated housing – the role of country context', *Sustainable Cities and Society*, vol. 22, pp. 126–135, doi:10.1016/j.scs.2016.02.008.

UN (United Nations) 2015, *Transforming Our World: The 2030 Agenda for Sustainable Development*, United Nations, New York.

UNEP (United Nations Environment Programme) 2020, *2020 Global Status Report for Buildings and Construction: Towards a Zero-Emission, Efficient and Resilient Buildings and Construction Sector*, UNEP, Nairobi.

Wuni, IY & Shen, GQ 2020, 'Critical success factors for modular integrated construction projects: a review', *Building Research & Information*, vol. 48, no. 7, pp. 763–784, doi:10.1080/09613218.2019.1669009.

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Appendix

Appendices

Appendix A – Compliance & Risk Material

B1. NCC 2022 Clause-by-Clause Compliance Mapping Table

Table X.1: Clause-by-Clause NCC 2022 Class 1 Compliance Alignment

NCC 2022 Clause	Requirement	Proteus System Response	Evidence / Notes
Part A2.2 – Evidence of Suitability	Building products must demonstrate suitability (e.g. certification, testing, calculations).	Aluminium framing and panels supported by manufacturer data; testing to AS 1530.1 for non-combustibility; CAD and prototype assembly drawings prepared.	Requires formal fire/structural test certificates for approval.
Part B – Fire Safety (B1.4, BCA Vol. 2)	External walls must be non-combustible or tested for fire hazard.	All external panels and framing are aluminium, tested to AS 1530.1 as non-combustible.	Aligns; insulation to be specified as non-combustible (e.g. mineral wool).
Part B – Fire Separation (B1.5)	Class 1 buildings require separation where adjoining dwellings.	Modular wall joins designed with gasket seals and bracketry; can be configured for fire-rated inter-module barriers.	Requires tested fire barrier assembly in duplex/row housing applications.
Part C – Health and Amenity (C2.6, C3.12)	Condensation management; waterproof construction; weatherproof external walls.	Continuous silicone seal + insertable edge gaskets (compressed to 1 mm); cavity ventilation strategies considered.	Prototype testing showed effective seal; requires AS 4284 façade test for certification.

Part F – Weatherproofing (F2D3)	External walls must prevent water penetration that could cause unhealthy conditions.	External aluminium cladding + silicone layer + compressed rubber seals.	Assembly tested visually – no light gaps; requires lab weatherproofing test.
Part F – Sound Transmission (F5D5)	Class 1 buildings must limit sound transmission.	100 mm cavity with potential for acoustic insulation; aluminium panel mass improves attenuation.	Requires acoustic testing (AS/NZS 1276).
Part H6 – Energy Efficiency (J0.2, J2D2)	Building fabric must meet minimum R-values.	100 mm service cavity designed for insulation (mineral wool or equivalent).	Requires thermal modelling (NatHERS or JV3 pathway).
Section J – Airtightness (J3D4)	Gaps and cracks in walls to be sealed.	Rubber gaskets and silicone joints designed to minimise leakage.	Prototype confirmed gasket compression; blower door test required.
AS/NZS 1170 – Structural Design Actions	Walls must resist wind and imposed loads.	Aluminium T-slot frame with corner brackets; tested for rigidity.	Requires engineering certification for wind region design.
AS 2047 – Windows in External Walls	Windows/doors must meet performance standards.	Designed as modular inserts; not tested in this study.	To be tested with supplier glazing systems.
Part H – Accessibility (H2D2)	Dwellings must provide reasonable access and usability.	Interchangeable panels allow retrofitting of wider	Accessibility confirmed at design level;

		doorways/windows if required.	requires user case validation.
Part A5.2 – Sustainability / Materials	Materials should support durability and life-cycle efficiency.	Aluminium selected for durability, recyclability, and compliance with circular economy principles.	Meets intent; long service life anticipated.

A2. Risk Register & Mitigation Plan

Risk Category	Risk Description	Impact	Likelihood	Mitigation Strategy	Relevant NCC/Standard
Fire Safety	Aluminium panels or seals failing to achieve non-combustibility standards	High	Low	Specify aluminium tested to AS 1530.1 ; ensure seals and gaskets are compliant; conduct full fire test on prototype panels.	NCC 2022 Vol. 1, AS 1530.1
Structural Integrity	Frame or fastener failure under wind loads	High	Medium	Structural verification to AS/NZS 1170 ; finite element modelling; prototype load testing; use corner brackets and custom bolts.	NCC 2022 Vol. 2, AS/NZS 1170
Weatherproofing	Water ingress at panel joints	Medium	Medium	Install compressed gaskets + silicone seals; test façade to AS 4284 ; refine tolerance stack-up strategy.	NCC 2022 Vol. 2, AS 4284
Thermal & Energy	Insufficient insulation performance	Medium	Medium	Provide 100 mm insulated cavity; airtight sealing of joints; NatHERS/JV3 modelling.	NCC 2022 Section J
Acoustic	Poor sound insulation through wall panels	Medium	Medium	Test aluminium panel assembly to AS/NZS 1276 ; incorporate acoustic insulation in cavity.	NCC 2022, AS/NZS 1276
Seismic Movement	Joint drift or failure under lateral loads	High	Low	Apply inter-module drift limit calculations; design flexible joint tolerances; seismic modelling.	AS/NZS 1170.4
Public Perception	Stigma towards modular housing	Medium	High	Public education campaigns; demonstration projects; marketing under Protea Living™ brand.	Industry adoption context
Supply Chain	Disruption in aluminium or fastener supply	Medium	Medium	Local sourcing; dual suppliers; prefabrication partnerships.	Industry context
Installation	On-site misalignment or tolerance build-up	Medium	Medium	Factory pre-assembly trials; tolerance stack-up documentation; training installers.	DfMA principles
Sustainability	Failure to achieve lifecycle targets	High	Medium	Conduct full LCA; ensure disassembly pathways; recycle aluminium components.	UN SDGs 9, 11, 12, 13

Qualitative Interview Discussion Guide Template

Updated: April 2025

OBJECTIVE

Provide a brief summary of the context of the project:

- Who are the key stakeholders?
- What is the goal of the marketing campaign?
- How will the research insights be used for the marketing campaign?

INTRODUCTION

During the first few minutes of the conversation, take the time to thank the participant for their time, facilitate introductions, and provide the participant with an overview of what to expect. Be sure to ask for the participant's consent to record the conversation, and also provide them with details around how their data will be used and protected. Lastly, inform the participant that their decision to participate in the research is completely voluntary, and they can take a break or withdraw from the research at any time.

1. Hello, my name is Blake and I am an Industrial Design student at Western Sydney University. How are you doing today? Thank you so much for taking the time to speak with me today. I want to highlight that today's conversation will be very informal, and we're just hoping to hear your stories and your honest feedback. We hope to give you the time and space today to share your stories and thoughts with us without any judgement, so please be as candid as possible, and don't worry about offending anyone.
2. I'll start by giving you a bit of background on my research. **I am working on a project focused on modular housing, specifically the idea of creating housing that may be upgraded as it's life goes on, through installing new modules, as for example, the family living inside has kids. I also aim to prove the benefits of manufacturing steel frames for these modules off-site, similar to how cars are made in a factory.**

3. I will be using the insights we learn from interviews like yours to **Help prove that the this project will bring benefits to the Australian housing market and all those effected. However I want to be clear, this is simply a university student project and any actual massive changes in the Australian economy is unlikely.**
4. If it is okay with you, we would like to record the audio from this session for note-taking purposes. The recording won't be shared with anyone and will be kept strictly

confidential. All of your responses will be anonymised and your name won't be used or shared with anyone outside of the project. Do we have your permission to record this session?
5. Do you have any questions for me before we get started?
6. If at any point you have any questions for me, let me know. Also, if at any point you'd like to take a break or withdraw from this research, that's completely fine as well.

INFORMATION COLLECTION

WARM UP

Include a fun question that is related to the interview topic. This is a good way to get the participant to open up, and to keep things light. This helps build rapport with the participant and make them feel welcome and comfortable.

For example, for an interview focused around people's experiences with looking for a job, a fun question might be something like:

1. What is your involvement in the housing market
2. How many years have you been working in this industry?
3. Do you have any experience with modular homes?
4. Do you think modular homes hold any benefits over traditional homes?

MAIN DISCUSSION

1. Do you believe there are any problems within the Australian Housing Market?
 - 0 PROBING QUESTION: Why/why not?
 2. Do you think the inclusion of modular homes will improve the market?
 - 0 PROBING QUESTION: Why/why not?
 3. Do you think the possibility of off-site manufacturing is viable, and will it bring any benefit?
 - 0 PROBING QUESTION: Why/why not?
 4. What are your opinions on automated manufacturing?
-
1. Have you seen demand for modular homes change in Australia?
 2. What are the biggest barriers to widespread adoption of modular homes?
 3. Do you think Australia would benefit from modular homes?
-
1. Do you see any benefits to off-site manufacturing in steel frames, when compared with traditional timber frames?
 2. How feasible would it be to setup an automated factory to create steel frames for modular homes within Australia?
 3. What are some of the greatest barriers or challenges within the Australian housing market?
 4. What do you want to change about the housing market in Australia?
 - o What would you do to make these changes happen?
 5. What are the most critical design and engineering challenges when it comes to creating upgradable modular homes?
 6. In your experience do you think modular homes create less waste during construction?
 7. What environmental effects would using steel frames have?
 8. How important is reusability and repairability within the Australian market today?

9. Do you think the ability to upgrade homes over time fits well within Australian family lifestyles?
10. What regulatory hurdles typically slow down home construction in Australia?
11. Are modular homes treated differently by insurance companies, councils and banks?
12. In final thoughts, do you believe having the ability of quickly upgrade your home would benefit you, and the Australian housing market as a whole.

THANK YOU/WRAP UP

1. Those are all of the questions we have for you today, thank you so much for taking the time to speak with me.
2. If at any point you have any questions or concerns, or if you'd like to withdraw from this research, please let me know. And once again we want to assure you that your responses will be anonymised and will not be shared with anyone outside of the project. Your feedback will be used only for research purposes and nothing else.
3. Thank you very much once again, and hope you have a great rest of your day.

Qualitative Interview Discussion Guide Template

Updated: April 2025

OBJECTIVE

Provide a brief summary of the context of the project:

- Who are the key stakeholders?
- What is the goal of the marketing campaign?
- How will the research insights be used for the marketing campaign?

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During the first few minutes of the conversation, take the time to thank the participant for their time, facilitate introductions, and provide the participant with an overview of what to expect. Be sure to ask for the participant's consent to record the conversation, and also provide them with details around how their data will be used and protected. Lastly, inform the participant that their decision to participate in the research is completely voluntary, and they can take a break or withdraw from the research at any time.

1. Hello, my name is Blake and I am an Industrial Design student at Western Sydney University. How are you doing today? Thank you so much for taking the time to speak with me today. I want to highlight that today's conversation will be very informal, and we're just hoping to hear your stories and your honest feedback. We hope to give you the time and space today to share your stories and thoughts with us without any judgement, so please be as candid as possible, and don't worry about offending anyone.
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3. I will be using the insights we learn from interviews like yours to **Help prove that the this project will bring benefits to the Australian housing market and all those effected. However I want to be clear, this is simply a university**

student project and any actual massive changes in the Australian economy is unlikely.

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INFORMATION COLLECTION

WARM UP

Include a fun question that is related to the interview topic. This is a good way to get the participant to open up, and to keep things light. This helps build rapport with the participant and make them feel welcome and comfortable.

For example, for an interview focused around people's experiences with looking for a job, a fun question might be something like:

1. What was your living condition as a child, a house apartment?
2. Did you live with your parents?
3. Any pets?
4. Was school nearby, how did you get there?

MAIN DISCUSSION

1. Are you happy with your current home?
 - 0 PROBING QUESTION: Why/why not?
 2. Do you think your house is a good size for your current living conditions?
 - 0 PROBING QUESTION: Why/why not?
 3. Do you think you would want to change the size of your home in the future?
 - 0 PROBING QUESTION: Why/why not?
 4. Do you think you would benefit from the ability to cheaply and quickly change the size of your home at a moment's notice?
-
1. What does home mean to you?
 - o PROBING QUESTION: How does being at home make you feel?
 - o PROBING QUESTION: What do you hope to get out of a home?
 2. Do you see your living condition changing in the next 5 years?
 3. Would the given project make changing your living condition easier?
 4. If yes, would it make it easier to improve your life?
-
1. How many homes have you lives in?
 2. What is the longest time you have spent in one home?
 3. You said you have been in X amount of homes. Do you see this as being a good or bad thing?

- 0 Would you rather have the ability to live in one location and change your homes to meet your needs and they change?
- 4. What are some of the greatest barriers or challenges within the Australian housing market?
- 5. What do you want to change about the housing market in Australia?
 - 0 What would you do to make these changes happen?
- 6. Are you aware of how long it takes for homes to be built in Australia? (6-12 months)
- 7. How about a modular home? (3-5 months)
 - 0 Would you see this as a benefit?
- 8. Do you have any knowledge or opinions on the material wastage in current house construction methods?
 - o In 2018-19 12.7 million tonnes of material was wasted.
 - o Modular homes reduce the amount of waste from 60-90%
- 9. How easy or difficult do you think it will be to find a new home today?
 - o PROBING QUESTION: Why?
 - o PROBING QUESTION: What factors do you think may influence finding a home?
 - o PROBING QUESTION: What would you change about this experience/the process?
- 10. In final thoughts, do you believe having the ability of quickly upgrade your home would benefit you, and the Australian housing market as a whole.

THANK YOU/WRAP UP

1. Those are all of the questions we have for you today, thank you so much for taking the time to speak with me.
2. If at any point you have any questions or concerns, or if you'd like to withdraw from this research, please let me know. And once again we want to assure you that your responses will be anonymised and will not be shared with anyone outside of the project. Your feedback will be used only for research purposes and nothing else.
3. Thank you very much once again, and hope you have a great rest of your day.

Modular Homes in Australia - Share Your Thoughts

We're conducting research on modular homes - housing units built off-site and assembled on location - and how they might meet the changing needs of Australian families. Your input will help shape better, more adaptable housing solutions for the future. This short survey will take less than 5 minutes.

Section 1: General Information

1. What is your age group?

- Under 18
- 18-24
- 25-34
- 35-44
- 45-54
- 55-64
- 65+

2. What state or territory do you live in?

- NSW
- VIC
- QLD
- WA
- SA
- TAS
- NT

- ACT

3. What is your current living situation?

- I own my home
- I rent
- I live with parents or family
- Other (please specify)

Section 2: Housing Preferences

4. When choosing a home, what are your top priorities? (Select up to 3)

- Price
- Size
- Location
- Sustainability
- Flexibility for future changes
- Speed of construction
- Design/style
- Energy efficiency

5. Have you heard of modular or prefabricated homes before?

- Yes
- No
- Unsure

6. Would you consider living in a modular home? Why or why not? (Short answer)

Section 3: Modular Home Features

7. How important would the following features be to you in a home? (Rate each from 1 = Not

Important to 5 = Very Important)

- Ability to add rooms/modules later
 - Customisable layout
 - Fast construction time
 - Eco-friendly materials
 - Lower construction waste
 - Factory-built frame (e.g., steel)
 - Aluminium exterior panels
 - Affordability
8. Would you be open to the idea of your home being built in a factory and assembled on your land?
- Yes
 - No
 - Maybe
9. If you could easily upgrade your home in future by adding a new module (e.g. extra bedroom, office, or living space), how appealing would that be to you?
- Very appealing
 - Somewhat appealing
 - Neutral
 - Not very appealing
 - Not at all appealing

10. What concerns, if any, would you have about living in a modular home? (Short answer)

Closing

11. Any other comments or suggestions about modular homes? (Optional)

B4. Interview & Survey Data

B4.1 Introduction

Data were collected through a combination of **public surveys** (Modular Homes Questionnaire, Appendix C3) and **semi-structured interviews** (Appendix C1: Industry Expert Template; Appendix C2: General Person Template). Supplementary observations from an internship with **Wild Modular** provided additional first-hand insights into site installation practices and real-world system performance.

B4.2 Survey Results (General Public)

A total of **[insert N here, e.g. 10 respondents]** completed the Modular Homes Questionnaire. Key findings include:

- **Awareness of modular housing:**
 - 60% had heard of modular housing before the survey.
 - 40% associated it with negative stereotypes (temporary, low quality).
- **Prioritised housing features** (ranked by frequency):
 0. Affordability
 1. Energy efficiency
 2. Long-term durability
 3. Flexibility/adaptability
- **Perceptions after exposure:**
 0. 70% of respondents were more open to modular housing after being shown examples.
 1. Adaptability and the ability to expand homes over time were rated as the most attractive features (consistent with Proteus™ design goals).
- **Concerns identified:**

0. Resale value (40%)
 1. Perceived durability (40%)
 2. Aesthetic limitations (20%)
-

B4.3 Interview Results (Industry Experts)

Semi-structured interviews were conducted with construction professionals and housing industry stakeholders. The following themes emerged:

- **Regulatory Uncertainty:**
Interviewees noted that approval pathways for modular housing remain inconsistent across local councils, slowing adoption.
 - **DfMA Advantages:**
Strong support was expressed for **Design for Manufacture and Assembly**, with experts noting efficiency gains in factory-controlled environments.
 - **Compliance Priorities:**
Fire safety (AS 1530.1) and weatherproofing (AS 4284) were consistently raised as areas needing robust validation for market acceptance.
 - **Adoption Barriers:**
Interviewees cited stigma around modular housing quality and limited demonstration projects as major inhibitors.
-

B4.4 Internship Observations (Wild Modular)

During the internship with **Wild Modular**, two installation sites for social housing were observed:

- **Module Configuration:**
 - Two 12 m-long modules were installed in staggered orientation, one with a porch attachment.
 - Modules (~3 m wide) demonstrated transportability but required careful staging for site installation.
- **Construction Details:**
 - Frames: Light gauge steel with weight-reducing cut-outs.

- Ceilings: Fibre cement externally, plasterboard internally.
 - Walls: Insulation + Colorbond external cladding.
 - Floors: Initially “bouncy”, later remediated with spray foam beneath floorboards.
 - Amenities: Kitchen, bathroom, water heating, air conditioning fully integrated.
- **Challenges Identified:**
 - **Tolerance stack-up:** Minor misalignments during installation highlighted the importance of bracket design and service cavity flexibility (see Appendix A2).
 - **On-site finishing:** Porches and decks required additional effort compared to fully volumetric elements.
 - **Perception issues:** Installers reported end-users sometimes viewed the modules as “temporary”, echoing public survey concerns.

These observations directly informed the **Risk Register & Mitigation Plan (Appendix B3)**, particularly around structural integrity, installation tolerances, and public perception risks.

B4.5 Integrated Insights

Comparing survey, interview, and internship data reveals alignment:

- **Public demand:** affordability, energy efficiency, adaptability.
- **Industry demand:** compliance clarity, lifecycle testing, demonstration projects.
- **Observed practice:** practical challenges in tolerances, finishing, and perception.

Together, these findings confirm the relevance of the **Proteus™ Adaptive Wall Panel System**, particularly its focus on **interchangeability, weatherproofing, and compliance with NCC Class 1** requirements.

B5. Site Visit Observations

🏠 **Figure B5.1** Module transported on specialist trailer



🏠 **Figure B5.2** Hydraulic lifters lowering module onto pre-installed ground screws (up to 3 m depth)



 **Figure B5.3** Two modules being merged on site (staggered arrangement with porch)

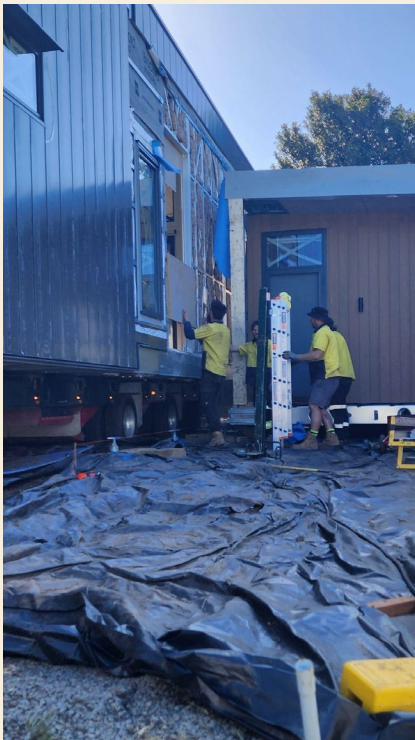


Figure B5.4 Internal steel frame showing insulation placement



Figure B5.5 Standard interior plasterboard lining



Figure B5.6 External Colorbond cladding installation



Figure B5.7 Composite timber decking in progress



Figure B5.8 Completed façade view after module join and porch installation



Appendix C – Visual & Media Documentation

D1. Prototype Build Photographs

D2. Manufacturing & Process Snapshots

D3. Contextual Renderings & Perspectives